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A Many-Body Dissipative Particle Dynamics Study of Forced Water–Oil Displacement in Capillary

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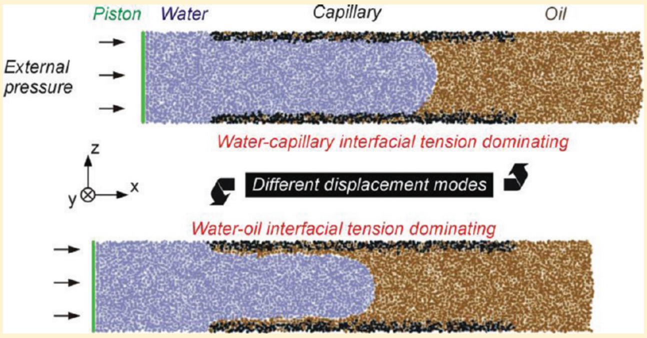
ABSTRACT: The forced water–oil displacement in capillary is a model that has important applications such as the groundwater remediation and the oil recovery. Whereas it is difficult for experimental studies to observe the displacement process in a capillary at nanoscale, the computational simulation is a unique approach in this regard. In the present work, the many-body dissipative particle dynamics (MDPD) method is employed to simulate the process of water–oil displacement in capillary with external force applied by a piston. As the property of all interfaces involved in this system can be manipulated independently, the dynamic displacement process is studied systematically under various conditions of distinct wettability of water in capillary and miscibility between water and oil as well as of different external forces. By analyzing the dependence of the starting force on the properties of water/capillary and water/oil interfaces, we find that there exist two different modes of the water–oil displacement. In the case of stronger water–oil interaction, the water particles cannot displace those oil particles sticking to the capillary wall, leaving a low oil recovery efficiency. To minimize the residual oil content in capillary, enhancing the wettability of water and reducing the external force will be beneficial. This simulation study provides microscopic insights into the water–oil displacement process in capillary and guiding information for relevant applications.



■ INTRODUCTION

The water–oil displacement in capillary is a simplified, yet realistic, model that mimics the microscopic situations in many natural and industrial processes, such as the groundwater remediation,[1] the oil recovery,[2] and nanofluidic devices.[3,4] In the case of groundwater remediation, the target is to remove the nonaqueous phase liquids (NAPLs), a class of hazardous organics insoluble in water, from the contaminated aquifer, while in the process of the oil recovery, the precious petroleum that has been trapped deeply in the field is expected to be completely extracted. Both processes share a common feature that the nonaqueous liquid (oil) in micropores (capillary) is going to be displaced by water, a process usually accompanied by external forces and also strongly related to the properties of the fluid/solid and fluid/fluid interfaces, such as the wettability and the miscibility.

The wettability of the fluid/solid interface and the miscibility of the fluid/fluid interface are two key properties that exert a great impact on the nature of the present system, such as the capillary pressure and the interfacial tensions, and mostly determine the dynamic behavior of displacement process.[5] To adjust these two parameters, surfactants have frequently been used, in particular in the oil recovery process. Albeit effective to some extent, the employment of surfactants is still far from optimal and rational. This is because, in addition to the complexity of reality, there has hitherto been no systematic understanding about how the involved interfacial properties and the external force influence the water–oil displacement process.

Although it is not difficult to perform in-situ observations on the displacement process in a capillary of diameter at micrometer scale, measurements for thinner capillaries (of nanoscale diameter) will be of great challenge. Moreover, it is difficult for experimental studies to disentangle the specific role of different interfaces in the present system because, upon applying a surfactant, all the interfaces are usually affected simultaneously. In other words, it is very difficult to tune a certain interface while freezing the others by means of experiment. For this purpose, the computational simulation is a more adequate approach, in which any interface in the system can be independently varied, and thus the mechanism of the water–oil displacement process in capillary can be unraveled in detail.

In the present work, we carried out a simulation study of the water–oil displacement in capillary with external force using the many-body dissipative particle dynamics (MDPD) method. By systematically analyzing the influences of the external force and individual interfacial tensions, we have gained important information about the starting pressure for displacement and the residual oil content (ROC); both are key parameters for the oil recovery process. We anticipate the data obtained from this simulation study and the resulting conclusion can provide insights into the water–oil displacement in capillary and guidelines for related applications.

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■ COMPUTATIONAL METHOD

Many-Body Dissipative Particle Dynamics. Since a detailed description of this method is presented in our previous research on spontaneous capillary imbibition and drainage,[6] we thus here give more general introduction of dissipative particle dynamics (DPD) and its many-body modification.

The DPD method, originally proposed by Hoogerbrugge and Koelman,[7] aiming at simulating flows at a coarse-grained level, has been demonstrated to be quite suitable for modeling the soft condensed matters such as simple liquids,[8] polymers,[9,10] and colloids[11,12] and other complex systems.[13] In comparison to the conventional molecular dynamics (MD), DPD allows for simulations at relatively larger scales of length and time, featuring the neglect of internal freedom degrees of atoms and molecules and the employment of weakly repulsive particle interaction.[14,15] According to our goals, direct simulation of system with capillary radius from nanoscale to microscale would bring classic MD unacceptable expensive burden, which facilitate the utilization of mesoscale modeling presented here.

Forces experienced by each DPD particle are divided into conservative, dissipative, and random parts: $\mathbf{F}_{ij}{}^{\mathrm{C}}$, $\mathbf{F}_{ij}{}^{\mathrm{D}}$, and $\mathbf{F}_{ij}{}^{\mathrm{R}}$, respectively. The conservative part is essentially a soft repulsive force, whereas the dissipative and random parts act as a thermostat which conserves the momentum and do not influence of the equilibrium.

Despite DPD method's versatility and its success upon being applied to various practical problems, it was pointed out by Louis et al.[16] that the quadratic equation of state (EOS) derived from the conservative force does not contain a van der Waals loop, which is quite different from the EOS of real fluid. Since the dissipative and random forces together act as a thermostat, which leaves the effective force between DPD particles solely repulsive due to the remaining conservative force, there is no force making the particles stick together; therefore, the simulation box tends to be fully filled with particles. Although DPD is adequate for studies on system with solely water/oil interfaces, it would be impossible to study systems with free surfaces under such conditions, especially in our studies involving water/vacuum and oil/vacuum interfaces.

Several groups[14,17,18] have tried to solve this problem by modifying the original DPD method to include the van der Waals loop in the EOS, which is referred as the many-body (or multibody) dissipative particle dynamics (MDPD). In the present work, we adopt the MDPD method reported by Warren,[13] in which the conservative part is modified as

$$\mathbf{F}_{ij}^{\mathrm{C}} = A_{ij} w_{\mathrm{C}}(r_{ij})\mathbf{e}_{ij} + B(\bar{\rho}_i + \bar{\rho}_j) w_{\mathrm{d}}(r_{ij})\mathbf{e}_{ij} \tag{1}$$

where A_{ij} is the maximum repulsion between particle i and j, $\mathbf{r}_{ij} = \mathbf{r}_j - \mathbf{r}_i$, $r_{ij} = |\mathbf{r}_{ij}|$, and $\mathbf{e}_{ij} = \mathbf{r}_{ij}/r_{ij}$. $w_{\mathrm{C}}(r_{ij})$ and $w_{\mathrm{d}}(r_{ij})$ are weight functions vanishing at distance beyond certain cutoff r_{c} or r_{d}.

$$w_{\mathrm{C}}(r) = 1 - r/r_{\mathrm{c}} \tag{2}$$

$$w_{\mathrm{d}}(r) = 1 - r/r_{\mathrm{d}} \tag{3}$$

The partial amplitude ρ depends on a weighted local density, and for the ith particle its value can be obtained as follows:

$$\bar{\rho}_i = \sum_{j \neq i} w_{\rho}(r_{ij}) \tag{4}$$

$$w_{\rho}(r) = 15(1 - r/r_{\mathrm{d}})^2/(2\pi r_{\mathrm{d}}{}^3) \tag{5}$$

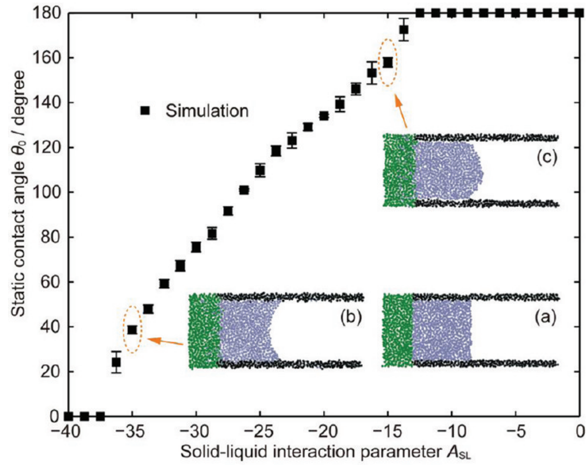

Figure 1. Relationship between the solid−liquid interaction parameter A_{SL} and the static contact angle θ_0 obtained by MDPD simulations. Insets are snapshots of the system in a common initial state before equilibration (a) and final states with $A_{\mathrm{SL}} = -35$ (b) and $A_{\mathrm{SL}} = -15$ (c) after equilibration.

In fact, the conservative force is divided into two individual parts: one is an attractive component obtained by simply turning the sign of the original force parameter, i.e., $A < 0$ and cutoff $r_{\mathrm{c}} = 1$, and the other is a many-body repulsive force with $B > 0$ and shorter cutoff $r_{\mathrm{d}} = 0.75$.

It has been demonstrated that with such modifications the theory-predicted shapes of both planar and curved liquid/vapor interfaces were successfully achieved.[14] Since it is our objective to clarify the general influence of several interfacial properties upon the final displacing process, we did not perform any mapping procedure for any component in MDPD as is done by Ghoufi et al. toward water.[19] Nevertheless, such work is of great importance for our later research focusing on specific combination of displacing and displaced fluids.

In the present study, the time step is chosen to be 0.01 MDPD time unit; the parameter B in eq 1 is kept 25 while A is set as −40 for interaction between the same kinds of particles, which leads to the equilibrium density of about 6.0. The tunable interaction parameter between water−capillary, oil−capillary, and water−oil are denoted as A_{W}, A_{O}, and A_{E}, respectively. All simulations are carried out in constant-energy, constant-volume NVE ensembles using the LAMMPS package,[20] and the MDPD algorithm is implemented by modifying the original DPD codes therein. Some of the simulation results are illustrated using the VMD software.[21]

■ RESULTS AND DISCUSSION

The Static State of Fluids in Capillary. To characterize the wetting properties of fluids within capillary, we conducted a series of calculations to correlate the static contact angle between a fluid and the capillary wall with the corresponding interaction parameter. In this calculation, a half-sealing capillary is employed and the whole processes are taken as follows: First, a 40 × 12 × 12 simulation box of a group of 26 609 homogeneous particles is equilibrated for 10^4 MDPD time units. Then particles at the left end of the box are defined as sealing slab, while the central particles in a cylinder region with a radius of 5 are defined as fluid particles, and remaining particles are set as capillary particles.

Table 1. Interaction Parameters for Static Contact Angle Calculations[a]

	capillary	fluid	slab
capillary	−40	A_{SL}	−40
fluid	A_{SL}	−40	−40
slab	−40	−40	−40

[a] A_{SL} is the solid−liquid interaction parameter to be varied in different simulations.

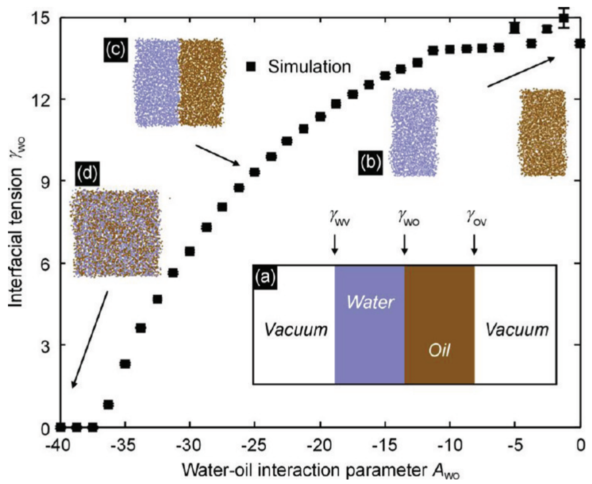

Figure 2. Relationship between the water−oil interaction parameter A_{WO} and the interfacial tension γ_{WO} obtained by MDPD simulations. Insets are the model used to calculate the γ_{WO} (a) and snapshots of system with A_{WO} = 0 (b), A_{WO} = −25 (c), and A_{WO} = −40 (d).

Finally, the right half of fluid particles is removed to leave a space for the formation of the meniscus (Figure 1a). The model proposed by Henrich et al.[22] is adopted to treat both the capillary wall and the sealing slab, in which particles are pinned to their pre-equilibrated positions by a spring force

$$\mathbf{F}_s = k_s(\mathbf{r}_i(t) - \mathbf{r}_i(0)) \tag{6}$$

with elastic coefficient k_s = 12.5. Interaction parameters involved in the static contact angle calculations are listed in Table 1.

In a NVE ensemble, 10^4 MDPD time units are left for a fluid to attain its stable meniscus shape and then 10^5 MDPD time units for data production. To obtain the contact angle, the fluid is divided into concentric cylindrical shells with thickness of $0.5r_c$, and the x position of each fluid/vacuum interface is exported by averaging over particles belonging; finally, a circular curve is used to fit the position data, and the contact angle is thus obtained.[6]

As shown in Figure 1, when the solid−liquid interaction parameter A_{SL} increases from −40 to 0, the wetting property of the fluid changes from completely wetting (θ_0 = 0°) to completely nonwetting (θ_0 = 180°), with partially wetting (Figure 1b) and partially nonwetting (Figure 1c) falling in between. This is in good agreement with our previous study[6] and similar simulations reported in the literature.[23]

The interfacial tension between water and oil is another crucial property that can be used to characterize the miscibility. The simulation system (Figure 2a) comprises a 5 × 10 × 10 water slab with 3040 particles and a 5 × 10 × 10 oil slab with 3043 particles attaching to each other. Two 35 × 10 × 10 vacuum regions are placed to the left and right of the fluids, leaving rich space for the formation of free water and oil interfaces. Interaction parameters involved in the interfacial tension calculation are listed in Table 2. Periodic boundary conditions are applied in three dimensions, mimicking infinite fluid layers and eliminating any undesired interference. 10^5 MDPD time units are left for equilibration and then 5 × 10^5 MDPD time units for data production.

Table 2. Interaction Parameters Involved in the Interfacial Tension Calculation[a]

	water	oil
water	−40	A_{WO}
oil	A_{WO}	−40

[a] A_{WO} is the water−oil interaction parameter to be varied in different simulations.

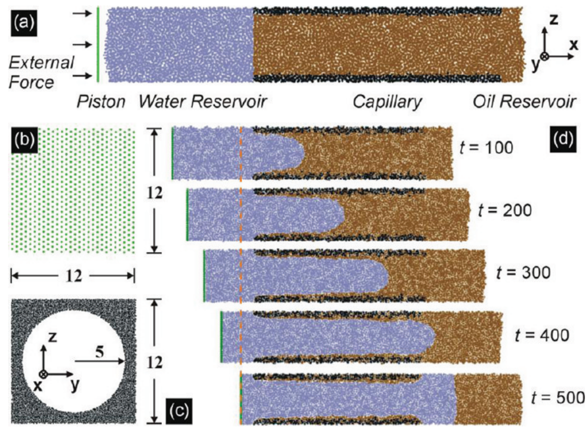

Figure 3. MDPD simulations for the water−oil displacement in capillary with external force. Water, oil, capillary, and piston particles are colored ice blue, brown, black, and green, respectively. (a) The initial state of system at t = 0 with external pressure applied to the left of the piston. (b) The piston layer with condensed particles. (c) The cross-section view of the capillary along the x direction. (d) Snapshots of the system at different time (A_W = −10, A_O = −40, A_E = −20, f = 3).

The total tension γ_{Total} of the system consists of three components (Figure 2a), i.e., water surface tension γ_{WV}, oil surface tension γ_{OV}, and water/oil interfacial tension γ_{WO}, as indicated in Figure 2a, and can be derived by subtracting mean tangential stress tensors (i.e., P_{yy} and P_{zz}) from the normal one (i.e., P_{xx}):[24]

$$\gamma_{Total} = L_x\left\langle P_{xx} - \frac{1}{2}(P_{yy} + P_{zz})\right\rangle \tag{7}$$

where L_x is the length of simulation box in the x direction. Since the γ_{WV} and the γ_{OV} remain unchanged, the γ_{WO} can be obtained by subtracting the two surface tensions from the total tension:

$$\gamma_{WO} = \gamma_{Total} - \gamma_{WV} - \gamma_{OV} \tag{8}$$

Different γ_{WO} can be calculated upon varying the water−oil interaction parameter A_{WO}, and the resulting relationship is shown in Figure 2. Apparently, with increasing the A_{WO} from −40 to 0 (the interaction between water and oil particles becomes weaker), the water/oil interface changes accordingly from being fully miscible to entirely immiscible. At the limit of A_{WO} = 0, the two immiscible fluids tend to detach from each

Table 3. Interaction Parameters Involved in the Water−Oil Displacement Simulations

	capillary	water	oil	piston
capillary	−40	A_W	A_O	−40
water	A_W	−40	A_E	−40
oil	A_O	A_E	−40	−40
piston	−40	−40	−40	−40

other (Figure 2b). In contrary, when A_{WO} reaches −40, the interface between two fluids will disappear, resulting in a fully miscible system (Figure 2d).

In general, successively altering the attraction parameter A while keeping repulsion parameter B unchanged (see eq 1) is a simple enough and effective approach to tune the interaction of concern. For the fluid−solid interaction, as A increase from −40 to 0 (see Figure 1), the attraction force decrease, resulting in the increase in the fluid/solid interfacial tension and a decline in the wettability of the fluid on the solid. For the fluid−fluid interaction, such a change can also cause the increase in the fluid/fluid interfacial tension and induce the decline in the miscibility.

Water−Oil Displacement in Capillary. Figure 3a depicts a typical simulation system of the water−oil displacement in capillary with external force. The capillary is similar in shape to that in the calculation of static contact angle (Figure 3c). A 120 × 12 × 12 box with 59 384 particles and periodic boundaries are equilibrated for 10^4 MDPD time units and then divided into four individual parts: water, oil, capillary, and piston. The capillary is designed to be prefilled with oil particles, and each side of the capillary is connected to a water and an oil reservoir on the left and on the right, respectively. Enough vacuum space (25 × 12 × 12) is left to both sides of the simulation box so as to eliminate the cross-boundary interactions. Interaction parameters involved in the water−oil displacement calculations are listed in Table 3.

To mimic the external pressure, we construct a piston layer consisting of condensed particles at the left-hand side of the simulation box (Figure 3b); forces and torques exerting on this layer will be summed up and constrained strictly along the x-axis. The water reservoir is designed to be large enough such that before the water front penetrates the right edge of capillary, the piston will not have touched the left edge of the capillary. The external pressure is calculated through dividing the total force by the cross-section area of the piston (Figure 3b). Specifically, the piston consists of 700 particles, and the cross-section area is 12 × 12; thus, 1 unit of force acting on each piston particle will in total bring 4.86 unit of external pressure on piston. For simplicity, we will discuss the effect of the external force, f, acting on each piston particle instead of using external pressure.

As the displacement commences, the external force acting on the piston will be transferred to the water particles, which are therefore driven into the capillary; meanwhile, the oil particles are displaced out of the capillary toward the oil reservoir at the right-hand side. Snapshots of a typical displacement process are displayed in Figure 3d. By counting the particle number of water and oil in the capillary at different time, the dynamic process of the displacement can be described in a more quantitative fashion, as shown by the resulting motion curves (the particle number−time relationship) plotted in Figure 4.

As indicated in Figure 4, the total number of fluid particles decreases slightly, which should be due to the decrease in the apparent density of fluids in the capillary during the displacement process. It is interesting to note that there is a linear region in both the water and the oil motion curves in Figure 4, which seems to suggest that the external force has effectively been compensated by the frictional force in the capillary during the water−oil displacement, resulting in the observed constant flow rate. In fact, such a linear change of the particle number with time cannot reveal the motion details of the water and the oil in the capillary. Upon entering the capillary, the water front keeps reshaping (Figure 3d), resulting in a variation in the dynamic contact angle of the water/oil interface during the whole displacement process. Therefore, although the water volume in the capillary seems to increase constantly with time, the water front moves, actually, with a small acceleration.

At the very beginning of the displacement process, the water particles surge into the capillary with relatively small resistance, thus resulting in a sudden increase in the water motion curve. While at the end of the displacement process, both the water and the oil motion curve take on a platform, suggesting that the water has penetrated the whole capillary, and the volumes of the water

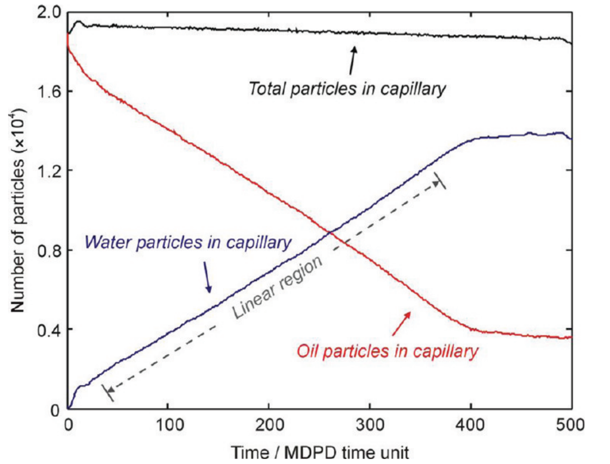

Figure 4. Change in particle numbers during the displacement process. Simulation parameters are the same as in Figure 3d.

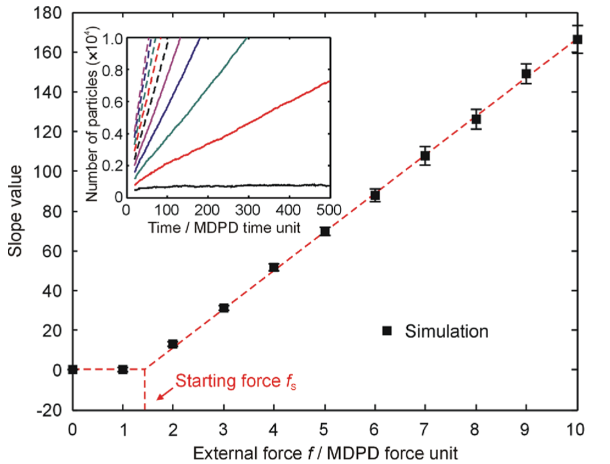

Figure 5. Slope of the linear region of water motion curves (shown as inset, the larger the external force the larger the slope) plotted against the external force. The starting force, f_S, is defined as the threshold force at the onset of this curve. Interaction parameters are taken as those in Figure 3d.

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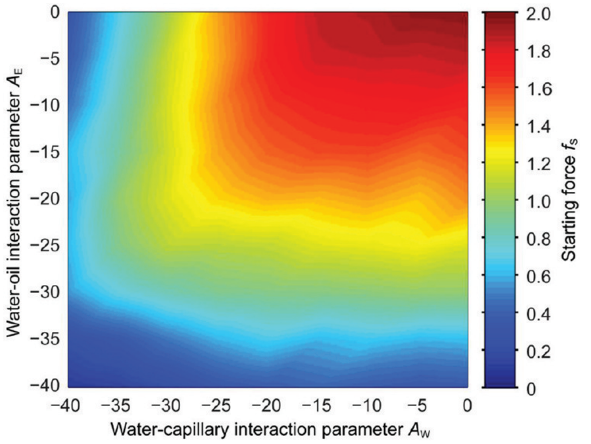

Figure 6. Dependence of starting force f_S on A_W and A_E (A_O is fixed at −40).

and the residual oil in the capillary remain approximately unchanged thereafter.

Starting Force. The slope of the linear region of the water motion curve (Figure 4) can serve as an index of the flow rate, which is a function of the interaction parameters (A_W, A_O, and A_E) and the external force (f). By freezing all the interaction parameters in MDPD simulations, the dependence of the flow rate on the external force can be obtained, as exemplified in Figure 5.

As revealed by Figure 5, the flow rate seems to increase approximately linearly with the external force in most of the range investigated, provided the external force has exceeded a threshold value, under which the water particles cannot enter the capillary. We thus define such a threshold force as the starting force, f_S. The existence of a starting force for the water−oil displacement in capillary should be due to the initial resistance of the system, which is an overall effect of all the involved interfacial tensions in the capillary and the wettability of both liquids. To simplify the following discussion, we only consider the most difficult case for the displacement process, i.e., to fix the oil−capillary interaction parameter A_O at −40 (corresponding an oil most sticky to the capillary) and to look at the dependence of the f_S on the other two interaction parameters, A_W and A_E.

Figure 6 illustrates how the f_S change with A_W and A_E. Colors are used to express the magnitude of f_S, with the dark blue and the dark red representing the lower and higher f_S, respectively. This figure comprises the $A_W \times A_E$ combinations of 9 × 9 cases, each with 10 simulations of different external forces to calculate the f_S; thus, 810 individual simulations in total are needed to plot Figure 6, from which the following two features can be found. First, the smallest f_S is found for systems taking the lowest A_W and A_E values, whereas the largest f_S is found for systems with highest A_W and A_E values. These results are understandable according to the Young−Laplace equation:[25]

$$\Delta p = \frac{2\gamma \cos \theta}{r} \tag{9}$$

where Δp is the pressure difference across the interface, γ is the interfacial tension, θ is the three-phase contact angle, and r is the radius of capillary. The validity of this equation for microscopic system is well demonstrated in previous simulation studies, e.g., spontaneous capillary imbibition and drainage.[22,23] On one hand, enhancing the wettability of water in capillary (i.e., lowering the A_W) will decrease the curvature of the water−oil meniscus and thus reduce the absolute value of cos θ at static state. On the other hand, lowering the A_E directly decreases the interfacial tension γ of the water/oil interface. Therefore, the f_S changes in the same tendency with both the A_W and the A_E.

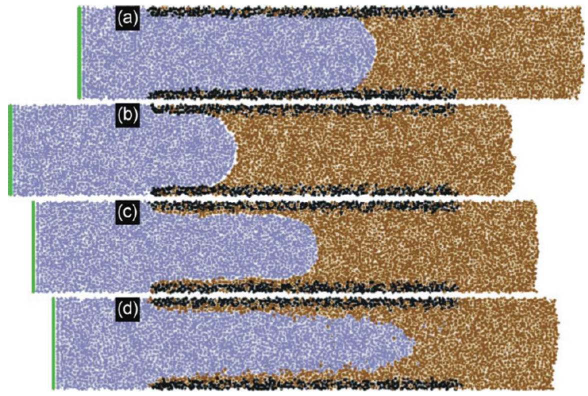

Figure 7. Two water−oil displacement modes. (a, b) are the A_w dominating mode, and (c, d) are the A_E dominating mode. Water, oil, capillary, and piston particles are colored ice blue, brown, black, and green, respectively. Simulation parameters: (a) A_W = −35, A_E = −20, f = 2, t = 500; (b) A_W = −20, A_E = −5, f = 2, t = 300; (c) A_W = −5, A_E = −20, f = 2, t = 500; (d) A_W = −20, A_E = −35, f = 2, t = 300. A_O is fixed at −40 in all the cases.

The second feature is a little subtle: when one of the two parameters (A_W or A_E) is fixed, the f_S will increase with the other parameter and then reach a limit value. For instance, when A_E is fixed at −20, the f_S increases with the A_W raised from −40 to −25 and then keeps almost unchanged even the A_W is further raised to 0. Similarly, when A_W is fixed at −30, the f_S will stay at a limit value over the range of A_E from −20 to 0. In other words, the color distribution in Figure 6 from the upper-right to the lower-left corner is not in a radial but in an approximately rectangular pattern.

Such a seemingly simple observation, however, unveils the in-depth mechanisms of the water−oil displacement process. The A_W and the A_E are, in fact, two factors determining the fashion how the water particles drive the oil particles. Specifically, when the A_W is fixed at a lower value relative to the A_E (i.e., the interaction between the water particles and the capillary wall is stronger), the water particles will take the place of all the oil particles near the capillary wall and push the whole oil cylinder to the right-hand side of capillary, as the cases of Figure 7a,b. When the A_E is fixed at a lower value relative to the A_W (i.e., the interaction between the water particles and the oil particles is stronger), the water particles cannot take the place of those oil particles at the capillary wall; the water−oil displacement process will occur in another fashion: the water particles will pierce the oil cylinder while leaving a significant part of oil particles still sticking to the capillary wall, as the cases in Figure 7c,d.

These two displacement modes are determined by the domination of A_W or A_E, which also explain the observed pattern of color distribution in Figure 6. When the water−oil displacement process is dominated by either A_W or A_E, further raising the other parameter (i.e., further reducing the corresponding interaction) will have little impact, leading to the fixed f_S effect observed in Figure 6.

ROC. Under the A_E dominating mode of water−oil displacement (Figure 7c,d), parts of the oil will remain in the capillary

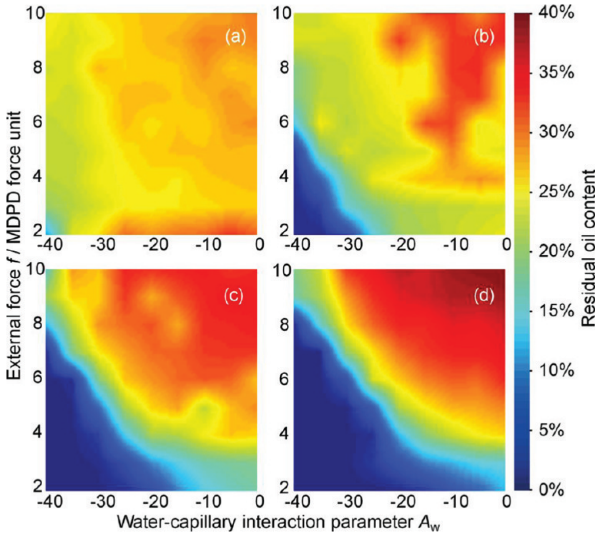

Figure 8. Change in the ROC upon varying f and A_W while fixing A_E at −35 (a), −25 (b), −15 (c), and −5 (d). A_O is kept at −40 in all cases.

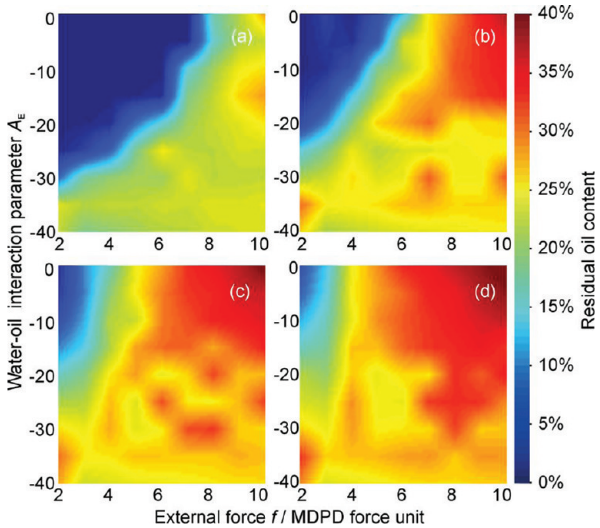

Figure 9. Change in the ROC upon varying f and A_E while fixing A_W at −35 (a), −25 (b), −15 (c), and −5 (d). A_O is kept at −40 in all cases.

even the whole capillary has been penetrated by water. The remaining percentage of oil in the capillary is defined as the residual oil content (ROC) in the present work, which is an important parameter describing the efficiency of the displacement. In the following, we shall discuss the influences of the external force f, the water−capillary interaction parameter A_W, and water−oil interaction parameter A_E on the ROC. Results are organized as Figures 8, 9, and 10. Colors are used to express the magnitude of ROC, with dark blue and dark red representing lower and higher ROC, indicating better and worse displacement efficiency, respectively. For each subplot in these figures, a data set obtained from 81 simulations are used to draw the color map.

Figure 8 presents the change in the ROC upon varying f and A_W while fixing A_E at −35, −25, −15, and −5, respectively. A general impression is that, at low A_E (e.g., −35), the ROC is relatively less sensitive to the variation of f and A_W (Figure 8a),

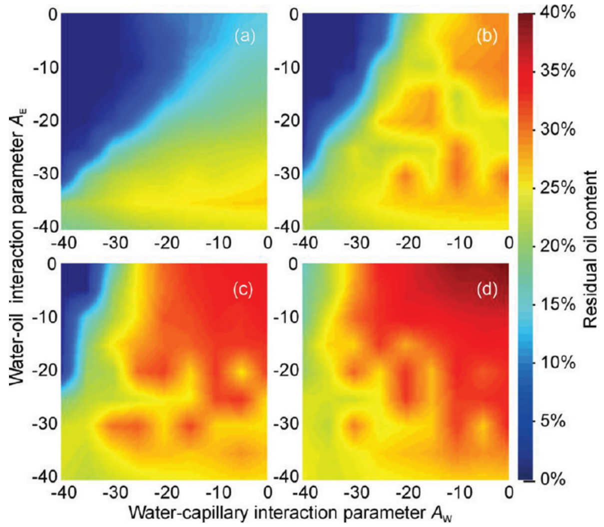

Figure 10. Change in the ROC upon varying A_W and A_E while fixing f at 3 (a), 5 (b), 7 (c), and 9 (d). A_O is kept at −40 in all cases.

while at high A_E (e.g., −5), the ROC change sharply with f and A_W, and smaller f and lower A_w will give better displacement efficiency.

Figure 9 illustrates the dependence of the ROC on f and A_E, with the A_W fixed at −35, −25, −15, and −5. Apparently, the lower the A_W, the smaller the ROC, as manifested by the blue color dominated Figure 9a and the red color dominated Figure 9d. This is in line with our previous discussion about two displacement modes, and the A_W dominating mode is clearly better for obtaining higher displacement efficiency.

Figure 10 plots the ROC as a function of A_W and A_E, with the f fixed at 3, 5, 7, and 9. The color distribution in the subplots of Figure 10 is very similar to that in Figure 9, manifesting that the smaller the external forces f, the less the oil remained in the capillary. Taking into account the displacement time, a moderate external force should be applied in reality to achieve optimal displacement efficiency in terms of the ROC and the consuming time.

Information conveyed by Figures 8−10 is, in fact, more than what have been briefly discussed above, and useful guidelines can be obtained for real applications aiming at increasing the oil recovery efficiency. Generally speaking, enhancing the water wettability in capillary and using small external force would be beneficial to the oil recovery efficiency. But the influence of the water−oil miscibility is not straightforward and depends on the specific combination of other parameters. For instance, in the case of small external force (e.g., Figure 10a), reducing the water−oil miscibility is always beneficial, while in the case of large external force (e.g., Figure 10d), sometimes enhancing the water−oil miscibility can have a better effect.

■ CONCLUSION

In the present work, we employ the MDPD method to simulate the water−oil displacement process in capillary with external forces applied by a piston. Through adjusting the interaction parameters, we are able to manipulate all the interfacial properties involved in this system, which enables us to define fluids with distinct wettability in the capillary and to tune the miscibility between water and oil.

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The dynamic process of the water−oil displacement in capillary can then be simulated with arbitrary combination of the water−capillary, oil−capillary, and water−oil interactions as well as with different external forces. We have focused on the most difficult case for water−oil displacement by setting maximum oil−capillary interaction. From the water motion curve we find that there exists a starting force (f_S) only beyond which can the displacement be triggered. By analyzing the dependence of f_S on the water−capillary interaction (A_W) and the water−oil interaction (A_E), two different displacement modes are revealed. In the A_W dominating mode, water particles can take the place of the oil particles at capillary wall and drive the whole oil cylinder out of the capillary, while in the A_E dominating mode, water particles can only drive the central part of the oil particles in capillary, leaving certain amount of oil sticking to the capillary wall, which is defined as the residual oil content (ROC).

To minimize the ROC, enhancing the water−capillary interaction and reducing the external force seem to be always beneficial, while the influence of the water−oil interaction is relatively complicated and depends on the specific combination of other parameters.

Our simulations provide microscopic insights into the mechanism of the water−oil displacement in capillary, which cannot be obtained by experimental studies, and gain useful guidelines for relevant applications such as groundwater remediation and oil recovery.

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REFERENCES

(1) Pankow, J. F.; Cherry, J. A. *Dense Chlorinated Solvents and Other DNAPLs in Groundwater: History, Behavior, and Remediation*; Waterloo Press: Portland, OR, 1996.

(2) Lake, L. W. *Enhanced Oil Recovery*; Prentice-Hall: Englewood Cliffs, NJ, 1989.

(3) Wang, W.; Yang, C.; Liu, Y.; Li, C. *Lab Chip* **2010**, *10*, 559.

(4) Liu, Y.; Wang, W.; Hu, W.; Lu, Z.; Zhou, X.; Li, C. *Biomed. Microdevices* **2011**, *13*, 769.

(5) Sahimi, M. *Rev. Mod. Phys.* **1993**, *65*, 1393.

(6) Chen, C.; Gao, C.; Zhuang, L.; Li, X.; Wu, P.; Dong, J.; Lu, J. *Langmuir* **2010**, *26*, 9533.

(7) Hoogerbrugge, P. J.; Koelman, J. M. V. A. *Europhys. Lett.* **1992**, *19*, 155.

(8) Groot, R. D.; Warren, P. B. *J. Chem. Phys.* **1997**, *107*, 4423.

(9) Groot, R. D.; Madden, T. J.; Tildesley, D. J. *J. Chem. Phys.* **1999**, *110*, 9739.

(10) Groot, R. D. *Langmuir* **2000**, *16*, 7493.

(11) Jury, S.; Bladon, P.; Cates, M.; Krishna, S.; Hagen, M.; Ruddock, N.; Warren, P. *Phys. Chem. Chem. Phys.* **1999**, *1*, 2051.

(12) Prinsen, P.; Warren, P. B.; Michels, M. A. J. *Phys. Rev. Lett.* **2002**, *89*, 148302.

(13) Shillcock, J. C.; Lipowsky, R. *Nature Mater.* **2005**, *4*, 225.

(14) Warren, P. B. *Phys. Rev. E* **2003**, *68*, 066702.

(15) Español, P.; Warren, P. *Europhys. Lett.* **1995**, *30*, 191.

(16) Louis, A. A.; Bolhuis, P. G.; Hansen, J. P. *Phys. Rev. E* **2000**, *62*, 7961.

(17) Pagonabarraga, I.; Frenkel, D. *J. Chem. Phys.* **2001**, *115*, 5015.

(18) Trofimov, S. Y.; Nies, E. L. F.; Michels, M. A. J. *J. Chem. Phys.* **2002**, *117*, 9383.

(19) Ghoufi, A.; Malfreyt, P. *Phys. Rev. E* **2011**, *83*, 051601.

(20) Plimpton, S. *J. Comput. Phys.* **1995**, *117*, 1.

(21) Humphrey, W.; Dalke, A.; Schulten, K. *J. Mol. Graphics* **1996**, *14*, 33.

(22) Henrich, B.; Cupelli, C.; Moseler, M.; Santer, M. *Europhys. Lett.* **2007**, *80*, 60004.

(23) Cupelli, C.; Henrich, B.; Glatzel, T.; Zengerle, R.; Moseler, M.; Santer, M. *New J. Phys.* **2008**, *10*, 043009.

(24) Jones, J. L.; Lal, M.; Ruddock, J. N.; Spenley, N. A. *Faraday Discuss.* **1999**, *112*, 129.

(25) Atkins, P.; de Paula, J. *Atkins' Physical Chemistry*, 8th ed.; Oxford University Press: Oxford, 2006.